# Acoustical Rainfall Analysis: Rainfall Drop Size Distribution Using the Underwater Sound Field

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ABSTRACT

Rainfall estimation is difficult, especially in oceanic regions where land-based techniques are unavailable. Fortunately, rain produces a loud and unique sound underwater that can be used to detect and quantify rainfall. Laboratory studies of the sound generated by individual raindrops have provided the basis for a formal inversion of the naturally generated underwater ambient sound field. Field measurements of subtropical rainfall at the Atlantic Oceanographic and Meteorological Lab Rain Gauge Facility are used to demonstrate the forward (predicting the sound field given the rainfall drop size distribution) and the inverse problem (estimating the drop size distribution given the sound field). This acoustical rainfall analysis (ARA) algorithm was tested for several dozen rainfall events spanning six months and was found to provide excellent estimates of rainfall rate, rainfall accumulation, and rainfall reflectivity (the quantity sensed by radars). High temporal resolution (order 5–10 s) variations in drop size distribution within the rain can be studied using ARA.

## 1. Introduction

Acoustical rainfall measurement uses the naturally occurring underwater sound produced by raindrops to quantitatively measure the rain. Components of acoustical rainfall analysis (ARA) include the detection, classification and quantification of rainfall. Detection is relatively simple. Rainfall produces a unique underwater acoustic sound that is easily distinguished from other common sound sources (breaking waves, biology, etc.) and is also much louder, by orders of magnitude, than these other sources. Unique sound signatures for different raindrop size have been identified (Medwin et al. 1992; Nystuen and Medwin 1995). Because of these different sound signatures, different rainfall drop size distributions produce distinct sound signatures, allowing classification of rainfall type. Finally, quantification is possible. The individual drop size signatures can be used to generate a mathematical basis with which the underwater sound field can be inverted to measure the drop size distribution within the rain. These measurements, in turn, can be used to calculate rainfall rate or other rain parameters, for example, reflectivity or liquid water content.

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## 2. Background physics

### a. The underwater sound of rain

Because the background noise field affects the performance of underwater acoustical systems, the underwater ambient noise field has been studied for decades (Knudsen et al. 1948). More recently, it has been recognized that the ambient sound field can be used as a signal to measure the processes generating the sound (Shaw et al. 1978; Nystuen 1986). In the frequency range from 1 to 50 kHz, the dominant sources of underwater sound are wind and rain (Wenz 1962). A summary of the mean sound spectra generated by wind and precipitation is shown in Fig. 1. The different spectral shapes for wind and precipitation are explained by studying the microphysics of wave breaking (e.g., Banner and Cato 1988; Medwin and Beaky 1989; Loewen and Melville 1994) and raindrop splashes (Pumphrey et al. 1989; Medwin et al. 1992; Nystuen and Medwin 1995). Because the shape of the wind and precipitation spectra are very different, each process can be acoustically detected and measured.

An example of the acoustic detection of light rain at sea is shown in Fig. 2. These data were collected using an autonomous acoustic drifter deployed off of the coast of California from May to August 1990. The drifter consisted of a hydrophone suspended 10 m beneath a small spar buoy. Acoustic data were collected once per hour and relayed to shore via the Argos satellite data link. Figure 2 shows the underwater sound spectrum before, during, and after the passage of a weak atmospheric front on 22 May. The shape of the

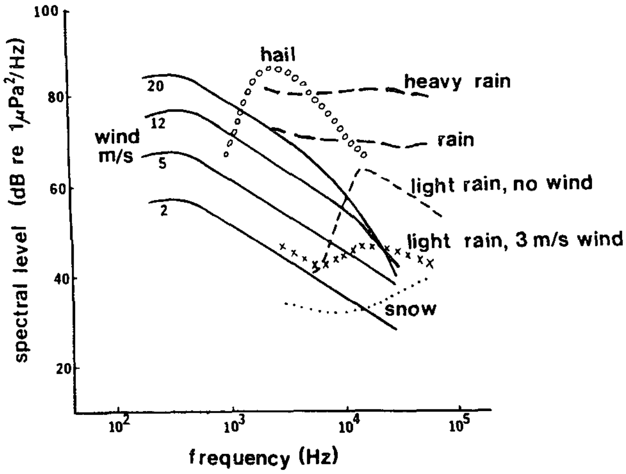


FIG. 1. Generalized oceanic ambient sound spectra for wind and precipitation. Note the distinctive spectral signatures for each phenomenon.

sound spectrum at 2019 UTC indicates drizzle (Nystuen 1993). As confirmation, a geostationary satellite image (GOES visible image) at 2031 UTC (Fig. 3) shows that the drifter was located underneath a cloud band associated with the atmospheric front (drizzle likely). The sound spectra prior to the front (at 1716 UTC) and after (at 2327 UTC) are typical of "wind only" generated underwater sound spectrum—that is, a relatively uniform spectral slope—that can be used to estimate wind speed. Using an acoustic wind speed algorithm (Vagle et al. 1990), the wind speeds before and after the passage of the front were 4.6 and 3.0 m s$^{-1}$, respectively.

An example of the acoustic detection and measurement of heavy rainfall at sea is shown in Fig. 4. Again, these data were reported from an autonomous acoustic drifter. This drifter was deployed in the Atlantic Ocean southwest of Bermuda in April 1992. On 29 April, a strong atmospheric front passed over the buoy location (35°N, 65°W). Three sound spectra are shown. The spectrum at 2206 UTC (28 April) and 2145 UTC (29 April), before and after the frontal passage, are typical "wind only" spectra. Using the acoustic wind algorithm (Vagle et al. 1990), wind speed estimates are 3.3 and 12.7 m s$^{-1}$, respectively. These estimates are in close agreement with near-simultaneous wind speed estimates, 3 and 11.5 m s$^{-1}$, respectively, from a passive microwave radiometer (SSM/I) onboard the Defense Meteorological Satellite Program (DMSP) F-8 satellite. At 0913 UTC, the SSM/I indicated "heavy" rain, 5 mm h$^{-1}$, over the location of the drifter. The acoustic estimate of rainfall rate at 0914 UTC, using the algorithm proposed by Nystuen et al. (1993), is 63 mm h$^{-1}$. The mismatch in rainfall rate estimates is not surprising. The SSM/I sensor has a spatial resolution of 50 km$^2$, and the SSM/I rainfall rate algorithm suffers from a lack of surface truth available for calibration.

## *b. Sound produced by individual raindrops*

The sound production mechanisms for individual raindrop splashes have been identified and quantified (Medwin et al. 1992). Sound is produced by raindrop impacts and oscillating bubbles trapped underwater during the raindrop splash. The impact sound depends on drop size, impact speed, and drop shape. Three mechanisms for trapping bubbles underwater have been identified. Type I bubbles are created by the unique crater geometry associated with 0.7–1.2-mm raindrops at normal incidence (Pumphrey et al. 1989). These small raindrops produce a conical splash crater that closes more rapidly from the sides (due to surface tension) than the bottom. A bubble of very predictable size is trapped at the apex of the splash crater. The resonant frequency of this bubble is 13–20 kHz. In light rain, containing large numbers of this drop size, the spectral character of the underwater sound shows a prominent peak at 13–20 kHz. Because the likelihood of bubble generation depends on incidence angle, the peak is sensitive to wind speed (Nystuen 1993).

Type II and type III bubbles are produced by the energetic splashes associated with large (2.4–4.1-mm diameter) and very large (over 4.1-mm diameter) raindrops (Nystuen and Medwin 1995). Type II bubbles are trapped underwater by a downward-moving turbulent jet associated with the splash canopy. They are created 50–80 ms after the initial impact and have resonant frequencies from 1 to above 30 kHz. Type III bubbles are trapped underwater by the reentry splashes of the aerosols thrown up by the initial impact. They occur 100–250 ms after impact and have resonant frequencies from 3 to above 30 kHz.

In summary, two raindrop sizes, small (0.8–1.1-mm diameter) and large (2.4– >5-mm diameter), are re-

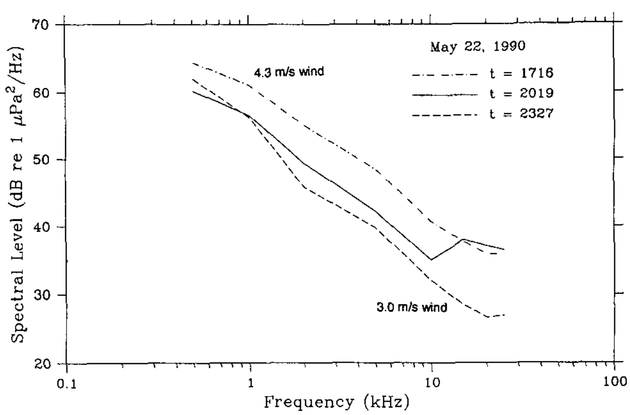


FIG. 2. Sound spectra recorded from an autonomous acoustic drifter during the passage of a weather front over the location of the buoy. The spectrum at 2019 UTC indicates light rain. The other two spectra are "wind only" spectra. The acoustic wind speed estimates using Vagle et al. (1990) are 4.6 (at 1716 UTC) and 3.0 m s$^{-1}$ (at 2237 UTC).

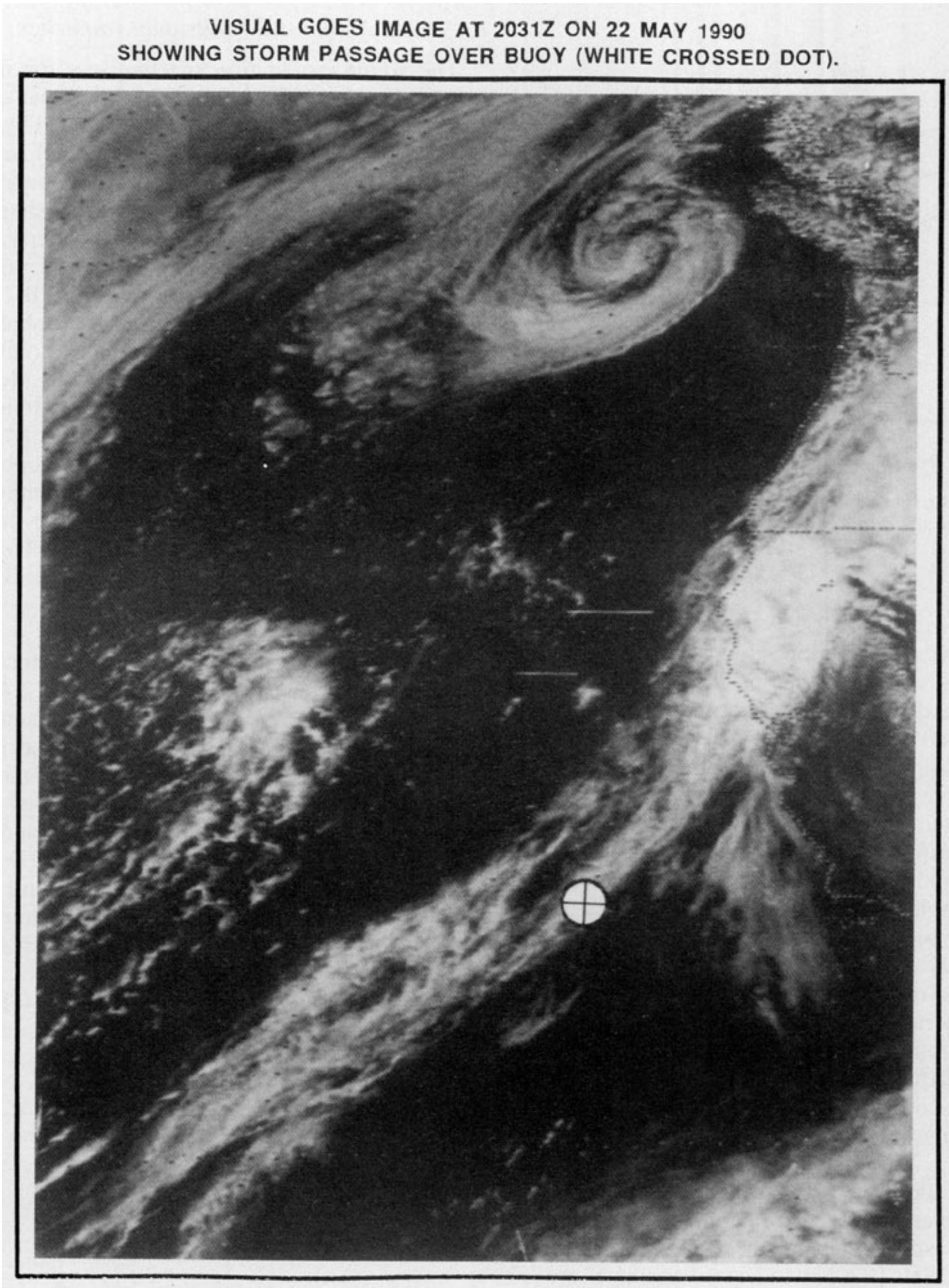


FIG. 3. Visual GOES image at 203 UTC 22 May 1990. The weather front is located over the buoy location,(indicated by the white crossed dot).

sponsible for most of the sound production by rainfall. Tiny drops (<0.8 mm) have not been shown to produce significant sound, and medium drops (1.1–2.4 mm) produce sound by a relatively weak process (the impact) compared to small and large raindrops. This understanding of the sound production mechanisms will be used to formulate the inversion matrix associating drop size with sound production.

## 3. Experimental description

### *a. The AOML Rain Gauge Facility*

The experiment took place adjacent to the Atlantic Oceanographic and Meteorological Lab (AOML) on Virginia Key, Miami, Florida (Fig. 5). Three rain gauge systems include the acoustical system, a Joss-Waldvogel RD-69 disdrometer, and an RM Young ca-

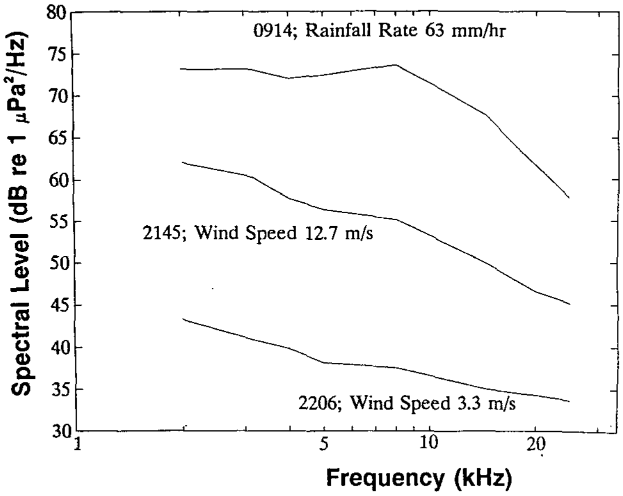


FIG. 4. Sound spectra recorded from an autonomous acoustic drifter near Bermuda on 28/29 April 1992. At 0914 UTC heavy rainfall is detected acoustically. At 0913 UTC, heavy rain was indicated in the vicinity of the drifter by a passive microwave radiometer (the SSM/I) onboard the DMSP F-8 satellite.

pacitance rain gauge. The acoustical system consists of an ITC-4123 hydrophone mounted in the center of a shallow, brackish water pond 1.5 m below the water surface and 0.5 m above the bottom. The pond has a soft mud bottom (low acoustic reflectivity, especially at the high frequencies used for acoustical rainfall analysis) and is surrounded by 5–10-m-tall mangrove trees. These trees provide some shelter from the wind, but are more than 30 m from the hydrophone location.

The hydrophone has an acoustic sensitivity of $-145$ dB rel 1 V $\mu Pa^{-1} \pm 1.5$ dB between 100 and 30 kHz. The signal is amplified and bandpass filtered (high pass at 300 Hz; low pass at 31.5 kHz). The signal is digitally sampled at 100 kHz (4096 points or 41 ms) once per second. These data have a Hanning window applied, a FFT performed, and then are averaged to produce one 64-point acoustic power spectrum with 86 degrees of freedom per second. The frequency resolution is 780 Hz. Further averaging can be performed to increase the degrees of freedom and to match the temporal sampling resolution of the disdrometer. Because of the high number of degrees of freedom associated with short time intervals, acoustical rainfall analysis can be used to study temporal resolution of rainfall not available to the other types of rain gauges.

One unique feature of the acoustical system is that the effective catchment area is the area of the water surface from which the sound generated by the rain propagates to the hydrophone. The individual sound sources, the raindrop splashes, are directional, vertically oriented acoustic dipoles (Medwin et al. 1992). This means that if the water is acoustically isotropic (uniform sound speed in all directions) and the rain is uniform across the surface, then 95% of the sound arriving at the hydrophone is coming from an area roughly equal to $\pi h^2$ directly above the hydrophone, where $h$ is the hydrophone depth (Nystuen et al. 1993). In the AOML Rain Gauge Facility, this corresponds to roughly 7 m$^2$, a cou-

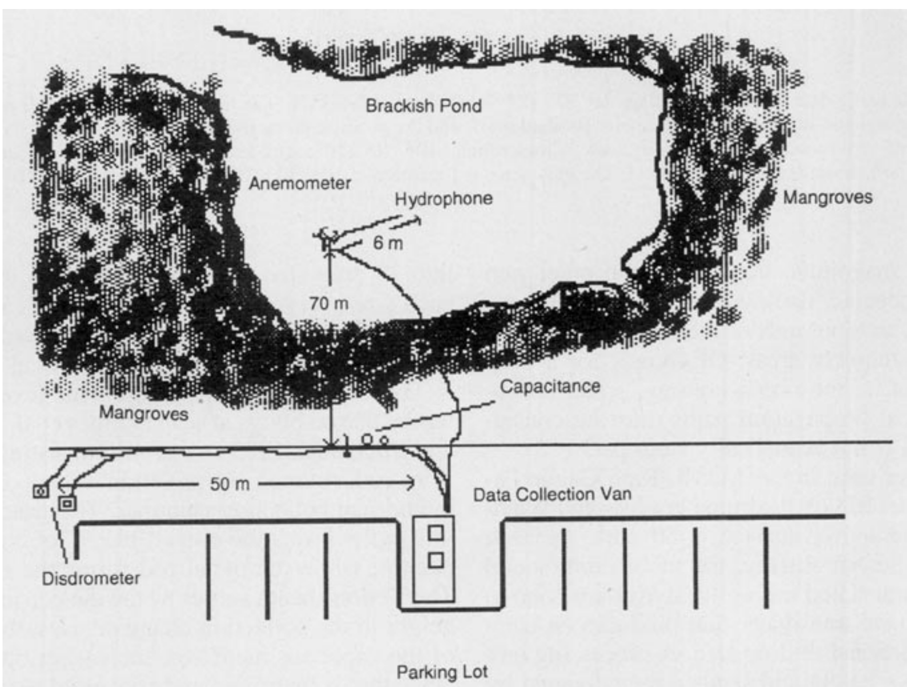


FIG. 5. Physical configuration for the experiment.

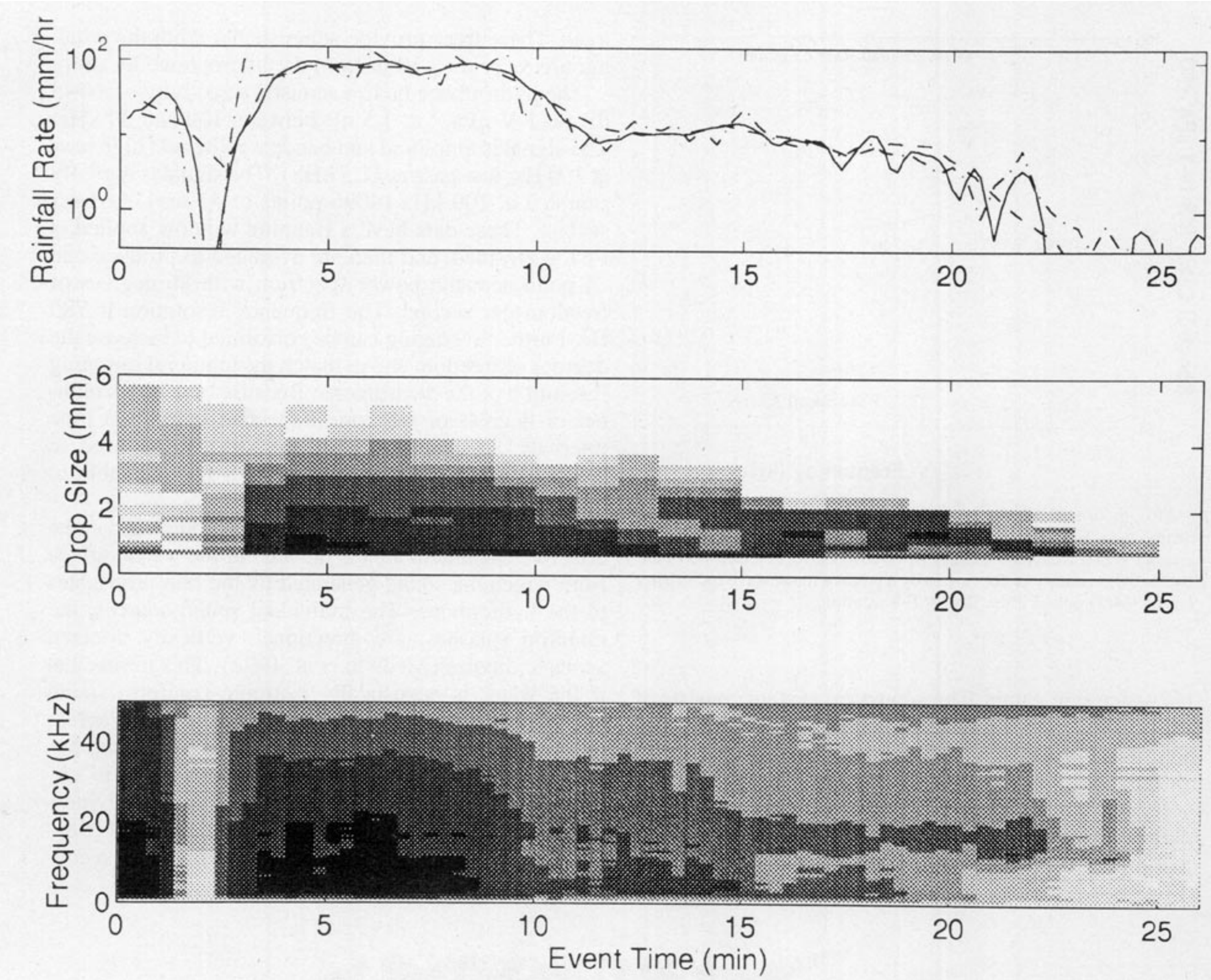


FIG. 6. An example using data collected on Julian day 272 (29 September) at 0233 LT. The top panel shows rainfall rate estimates from the capacitance rain gauge (solid line), the disdrometer (dashed line), and the acoustic method described in this paper (dash–dot line). The middle panel shows the disdrometer data. The gray scale is logarithmic: $10^0$, $10^1$, $10^2$, $10^3$, and $10^4$ drops per cubic meter per millimeter. The bottom panel shows the underwater sound field. The gray scale is logarithmic: $10^3$, $10^4$, $10^5$, $10^6$, $10^7$, $10^8$, and $10^9$ $\mu Pa^2\ Hz^{-1}$.

ple of orders of magnitude larger than the other rain gauges. In an oceanic deployment, the hydrophone depth can be tens to thousands of meters, allowing very large effective catchment areas. Of course, for a deep deployment—that is, the ocean bottom—careful consideration to actual propagation paths must be considered as the ocean is not acoustically isotropic.

The disdrometer used in the AOML Rain Gauge Facility is a Distromet RD-69 disdrometer (Joss and Waldvogel 1969). The sensor surface is 50 cm$^2$. As each raindrop hits the sensor surface, the momentum/sound of the impact is translated through a styrofoam cone to an electromechanical transducer that produces an electrical pulse proportional to drop size. A processing unit measures the pulse height and sends a digital signal by serial port to a computer. The computer bins the data into 20 drop size categories and stores drop counts for each size every minute. The drop size categories range from 0.3- to 5.5-mm diameter, although the smallest three bins are not reliable (Nystuen et al. 1996).

The capacitance rain gauge was developed for potential use on buoys at sea (Holmes et al. 1981; Holmes and Michelena 1983). A probe consisting of a stainless steel rod covered by a Teflon sheath is set inside a cylindrical collection chamber. The water surrounding this probe forms the outer "plate" of coaxial-type capacitor, while the metal rod forms the inner "plate." The Teflon sheath serves as the dielectric. As the water height in the collection chamber rises, the surface area of the capacitor increases, increasing the total capacitance that is measured and converted to an analog voltage directly proportional to the height of the water in

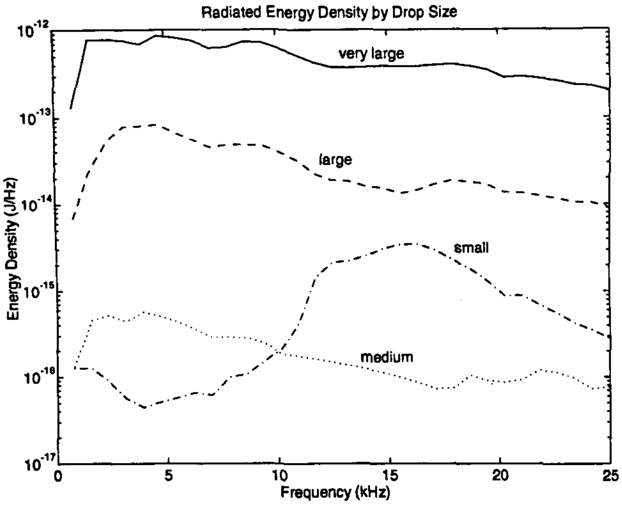


FIG. 7. The radiated energy density per raindrop for four drop size categories. The categories are small (0.7–1.2-mm diameter), medium (1.2–2.4 mm), large (2.4–4.1 mm), and very large (>4.1 mm diameter) raindrops.

the collection chamber. The measurement of rainfall rate is the difference in rainwater accumulation over a given time interval.

The capacitance rain gauge used in the AOML Rain Gauge Facility is an RM Young model 50202 precipitation gauge, based on the Holmes et al. (1981) design. This instrument has a catchment collection area of 100 cm$^2$. The capacity of the collection chamber is 50 mm of rainwater. The precision for rainfall accumulation using the 16-bit A/D board is 0.0015 mm of rain per bit. In practice, the flow characteristics of draining from the catchment collection area into the measurement chamber control measurement accuracy. To remove the influence of flow irregularities a 100-s binomial filter is applied to the data. The resulting 1-min rainfall rate resolution is roughly $\pm 1$ mm h$^{-1}$, although isolated errors of up to $+10$ mm h$^{-1}$ are still possible. The capacitance rain gauge has excellent performance characteristics when compared to weighing, tipping bucket, and optical scintillation rain gauges for rainfall rates over 5 mm h$^{-1}$ (Nystuen et al. 1996).

*b. Data collection*

Data collection is automated. The nonacoustic instruments are monitored every 10 s. Whenever a designated rain gauge indicates a rainfall rate over 1 mm h$^{-1}$ the system automatically "turns on." Acoustic sampling is started and all instruments are monitored once per second. Sampling continues to the end of the rainfall event (defined as 5 min after the rainfall rate has dropped below 1 mm h$^{-1}$). For this paper, 40 events with maximum rainfall rates over 10 mm h$^{-1}$ between September 1993 and April 1994 were evaluated. Of these events, 12 had maximum rainfall rates above 100 mm h$^{-1}$ (extremely heavy rain).

An example of rainfall rate, disdrometer, and sound field data is shown in Fig. 6. This particular rain event, on Julian day 272 (29 September), showed distinctive changes in the drop size distribution data that were closely associated with changes in the underwater sound field.

The top panel shows the total rainfall rate estimates during the event. The event starts with a couple of minutes of heavy rain ($R$ = 20–30 mm h$^{-1}$), followed by a couple of minutes of almost no rain. The main downpour ($R \approx$ 100 mm h$^{-1}$) lasts 6–7 min. The downpour is followed by stratiform drizzle lasting roughly 15 min.

The disdrometer data (middle panel) shows that this event started with small populations of large and very large raindrops (minutes 1–3). The strong convective rain cell, with rainfall rates near 90 mm h$^{-1}$, lasted from minutes 4 to 10. Following the convective cell, drop size sorting typical of subtropical stratiform rain is detected (minutes 11–25). From minutes 11 to 14, the rain contains large (2.4–4.1 mm), medium (1.1–2.4 mm), and small (0.8–1.1 mm) raindrops. From minutes 15 to 20, the rain contained medium and small raindrops. From minute 23 to 24, only small raindrops are present.

The acoustic data (bottom panel) shows that changes in the underwater sound field are closely associated with the changes in the drop size distribution. When the very large raindrops (>4.1 mm) are present (minutes 1–2), the sound levels are very high, ranging from nearly $10^9$ $\mu$Pa$^2$ Hz$^{-1}$ at 5 kHz to $10^7$ $\mu$Pa$^2$ Hz$^{-1}$ at 50 kHz, although the rainfall rate is only 20 mm h$^{-1}$. During the main convective cell (minutes 4–10), the sound levels are still high and relatively uniform, although the sound level above 40 kHz is reduced relative to the first minutes of the rain. Later, during the stratiform rain (minutes 11–24), a pronounced peak is present between 13 and 20 kHz. This is the acoustic signature of small (0.8–1.1 mm) raindrops. At times when the rain contains only small and medium raindrops (minutes 15–24), this spectral peak is especially evident. When large and very large raindrops are present in the rain, the 13–20 spectral peak is obscured (or suppressed?). Because subtropical stratiform rain contains few large and very large raindrops, this feature of the underwater sound field can be used to detect and distinguish between convective and stratiform rainfall. When only small raindrops are present in the rain, the sound levels below 13 kHz are at nonrain event levels—that is, no sound below 13 kHz is generated by rainfall containing only small raindrops. Finally, at the end of the event (minute 25), very low background levels, roughly $10^3$ 1 $\mu$Pa$^2$ Hz$^{-1}$, are observed from 1 to 35 kHz. These sound intensities are nearly six orders of magnitude lower than at the start of the rain event.

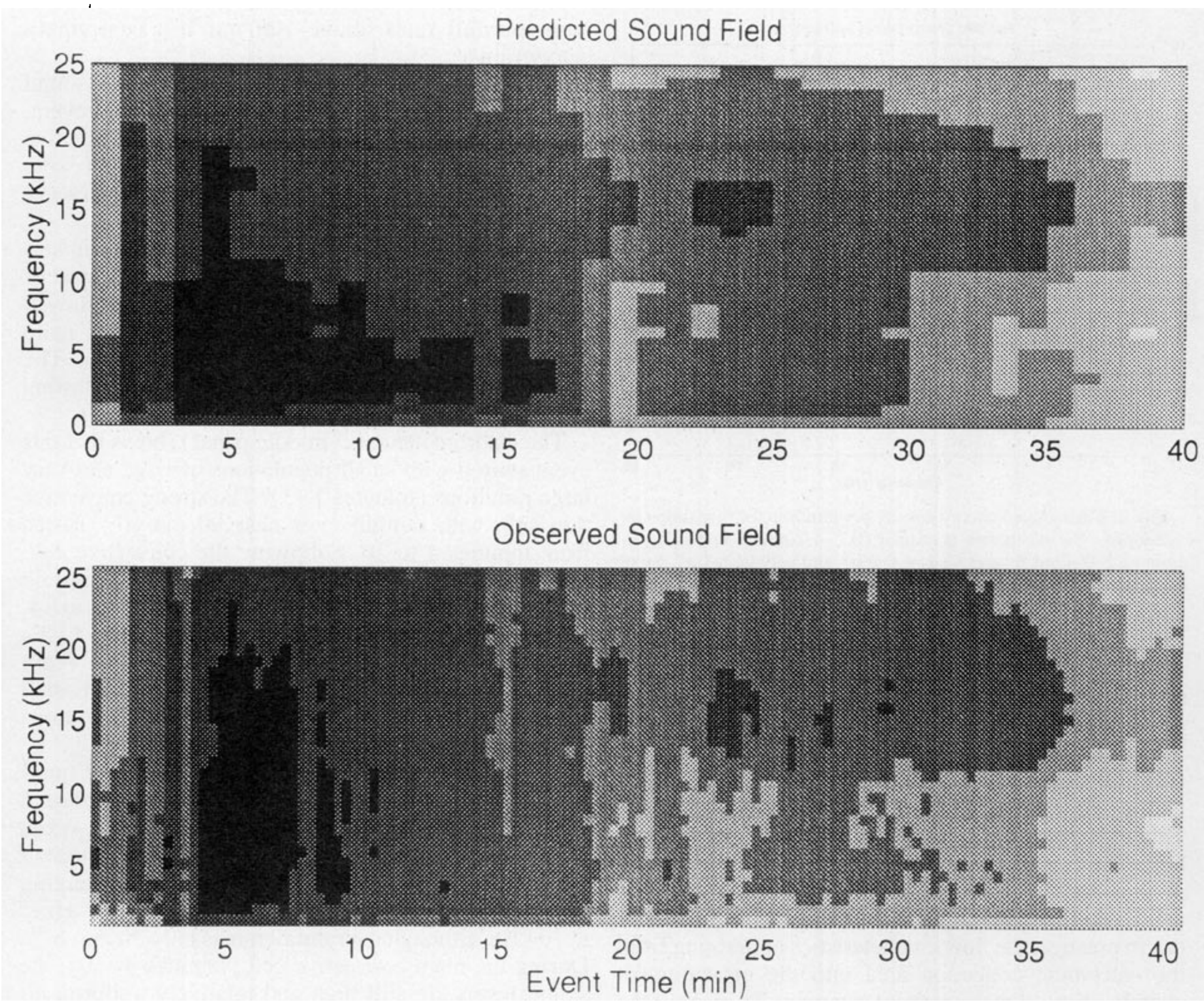


FIG. 8. Observed (bottom panel) and predicted (top panel) underwater sound fields during event 275 (2 October) at 0909 local time (LT). The gray scale is logarithmic: $10^4$, $10^5$, $10^6$, $10^7$, $10^8$, and $10^9$ $\mu$Pa$^2$ Hz$^{-1}$. The temporal resolution of the observed field is 20 s. The temporal resolution of the predicted field, based on the disdrometer data, is 1 min.

## 4. Data analysis

### *a. Inversion theory*

The forward problem, predicting the sound field given the drop size distribution, is followed by the inversion problem, predicting the drop size distribution given the sound field. The relationship between the underwater sound field and the drop size distribution in the rain is given by

$$I(f) = \mathbf{A}(f, D)\mathrm{drd}(D),$$

where $I$ is the underwater sound intensity, drd is the drop rate density (number of drops per square meter per second per millimeter), which is related to drop size distribution dsd (number of drops per cubic meter per millimeter) by a factor of terminal fall velocity, and a transformation matrix **A**, which contains the contribution to each frequency band $f$ by individual drops of size $D$. Matrix **A** also contains integration constants associated with the geometry of sound propagation from a surface sound source (assumed to be uniformly distributed vertical dipoles) and the hydrophone.

Matrix **A** is estimated empirically using disdrometer and acoustic data collected during the convective rainfall event on Julian day 272 (Fig. 6). At the end of the rain (minutes 23–24), only small (0.8–1.1 mm) raindrops are present. Using the small raindrop population measurement from the disdrometer, the mean contribution to the underwater sound field per small raindrop is estimated. Earlier in the event (minutes 15–21), only small and medium drops are present. Given the signature of a small raindrop and the drop population

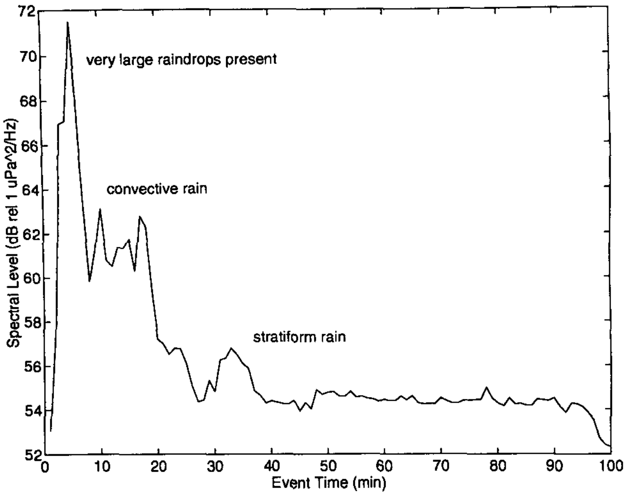


FIG. 9. The mean sound intensity level between 43 and 50 kHz during event 275 at 0909 LT. The absolute sound level in this relatively high frequency band appears to be a good indicator of rainfall classification.

measurements, the mean contribution per medium raindrop is estimated, and so on. Spectral signatures for four drop size categories were obtained: small (0.7–1.2-mm diameter), medium (1.2–2.4 mm), large (2.4–4.1 mm), and very large (>4.1 mm) (Fig. 7). These empirically derived acoustic signatures for each raindrop size agree qualitatively with the laboratory studies of single-drop splashes (Nystuen and Medwin 1995), although the spectra are more uniform and the levels are generally higher by 3–10 dB. The laboratory estimates of underwater sound signatures for individual drop sizes contain spectral shape fluctuations thought to be due to the relatively small number of individual drop splashes studies, roughly 400 drops. In a field deployment, many thousands of drops strike within the listening radius of the sensor, suggesting that the more uniform spectra are the result of averaging the sound produced by many more drops. The higher levels may be an indication of acoustic reverberation in the shallow pond environment. Alternatively, surface roughness due to drop splashes in natural rain may allow raindrops to create more bubbles, and consequently produce more sound underwater, than was observed in the laboratory studies (which generally used flat water surfaces).

The acoustic signatures shown in Fig. 7 closely parallel the laboratory results (Medwin et al. 1992). The large drops produce relatively uniform sound across the spectrum with a maximum near 3–5 kHz. Very large drops produce a similar spectrum, although louder and with a maximum level that extends to a lower frequency, 1.5–5 kHz. Small raindrops produce significant sound between 13 and 20 kHz. The only surprise when compared to the laboratory studies is the sound levels produced by the medium raindrops. In the laboratory studies, medium-sized raindrops have not been observed to generate underwater bubbles (Medwin et al. 1992) and, consequently, were thought to generate sound via the much weaker impact mechanism only. The levels shown in Fig. 7 are much higher than expected, although they are orders of magnitude lower than the other drop sizes reported.

Once the transformation matrix (matrix **A**; Fig. 7) is identified, the forward problem can be verified. This verification, using a different rain event (on Julian day 275), is shown in Fig. 8. All features of the observed sound field are predicted. At the start of this convective event (minutes 4–5), large and very large raindrops are present, producing loud and relatively uniform sound levels from 1 to 25 kHz. During the main convective cell, with extremely heavy rainfall ($R \approx 100$ mm h$^{-1}$), the very large drops are absent (minutes 6–18). The sound levels are still relatively uniform, but not as loud as earlier, especially below 2 kHz. Later, during the stratiform periods (after minute 19), the distinctive peak between 13 and 20 kHz due to small drops is present.

Once the solution to the forward problem has been established, the inverse problem is given by

$$\mathbf{A} = \mathbf{U}\Lambda\mathbf{V}^{\mathrm{T}},$$

$$\mathrm{drd} = \mathbf{V}[\Lambda^{-1}(\mathbf{U}^{\mathrm{T}}\mathbf{I})],$$

where **U**, Λ, and **V** are found by singular value decomposition of matrix **A**. Successful inversion requires that the basis for the inversion (the four drop size signatures

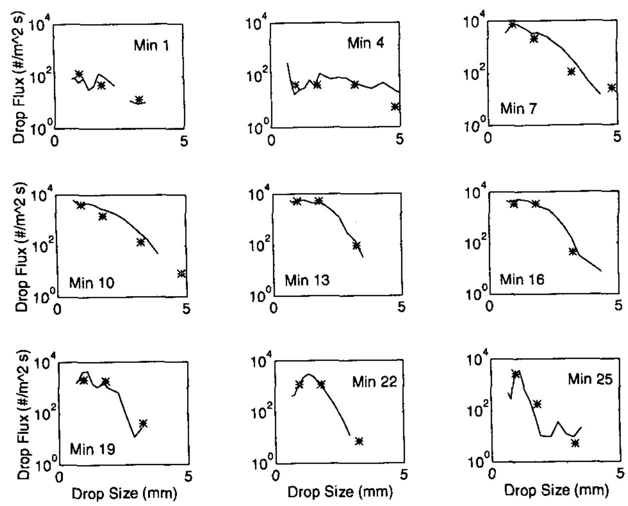


FIG. 10. Examples of the drop size distribution measurements for event 275 (2 October) at 0909 LT. The acoustic inversion values are indicated by the asterisk symbols; the disdrometer measurements are shown by the solid lines. Very large raindrops are acoustically detected in minutes 4, 7, and 10. The heavy rainfall occurs from minute 7 to minute 19. The stratiform rain occurs after minute 19 and shows a peak in the drop distribution near 1-mm drop diameter.

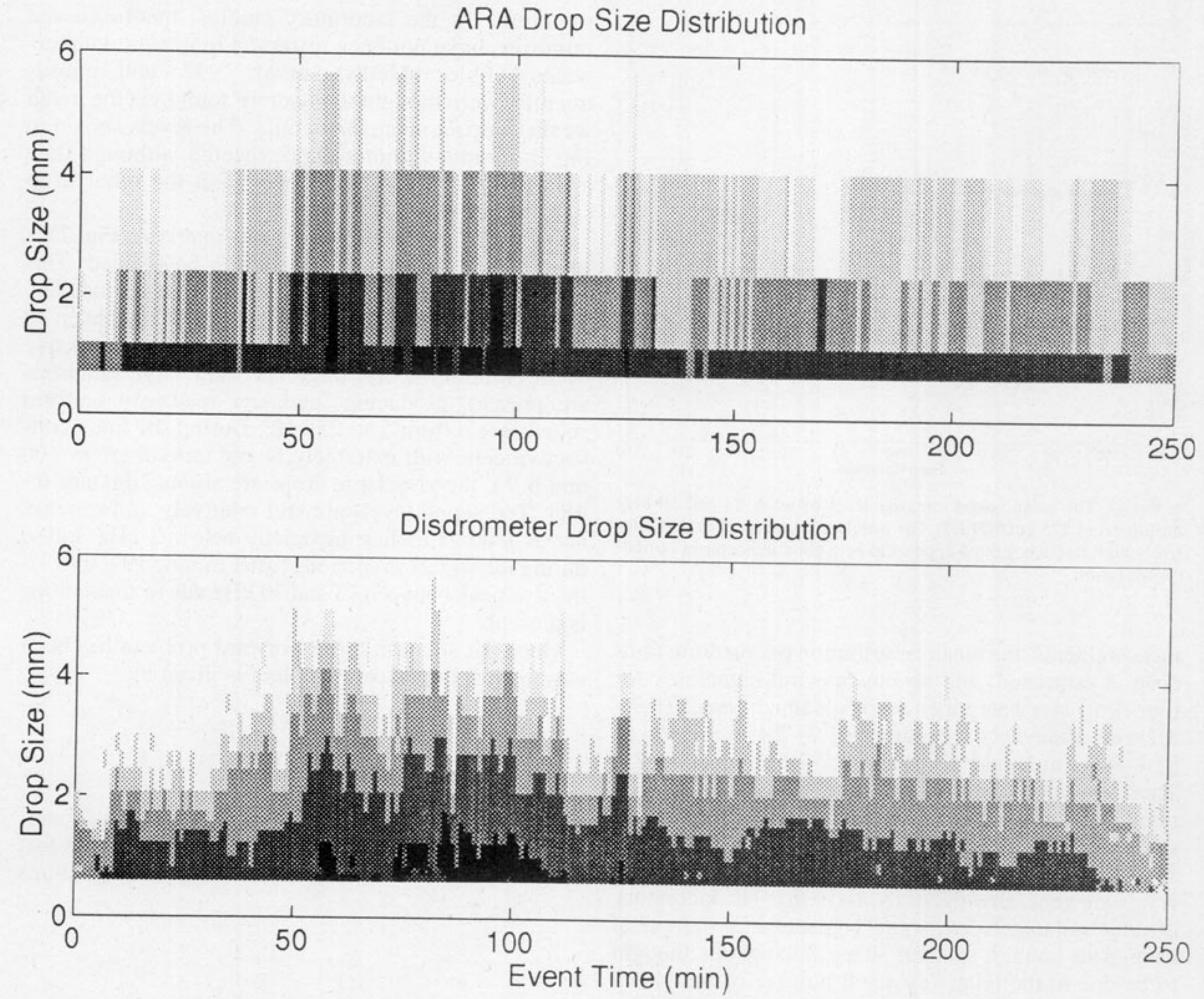


FIG. 11. Drop size distribution estimates for event 289 (16 October) at 0415 LT. The top panel shows the acoustic inversion (ARA) of the sound field. The bottom panel shows the disdrometer measurements. The gray scale is logarithmic: $10^{-1}$, $10^{0}$, $10^{1}$, $10^{2}$, and $10^{3}$ drops per cubic meter per millimeter.

in Fig. 7) have unique spectral shapes. In Fig. 7, the spectral shape of the very large and medium drop sizes are so similar, although different by orders of magnitude in amplitude, that the inversion can confuse a few very large raindrops with thousands of medium raindrops. Fortunately, an independent acoustic signal in the frequency band 43–50 kHz was found to be associated with the presence of very large raindrops (Fig. 9). Whenever very large raindrops were present, the sound level was observed to be above 63 dB rel 1 $\mu$Pa$^2$ Hz$^{-1}$ in this frequency band. This feature of the underwater sound field allows the rain event to be further sorted acoustically (convective rain with very large raindrops, convective rain without very large raindrops, and stratiform rain).

Two versions of the acoustic inversion were performed, each using only three of the basis vectors shown in Fig. 7. If very large raindrops were detected, the medium raindrop population was estimated using an interpolation between the small and large raindrop populations—that is, the medium drop signature was not used. If no very large raindrops were detected, then the inversion did not use the very large raindrop spectral signature. A final constraint on the inversion was that the drop size distribution spectrum decrease with increasing drop size.

Figure 10 shows examples of the acoustic inversion to obtain drop size distribution with the disdrometer data for the rain event on 2 October 1993 (the same event that was shown in Figs. 8 and 9). Although only four drop size populations are estimated using the inversion, most of the features of the rain events are detected. Accurate population estimates of very large raindrops occur on minute 4. The stratiform rain begins at minute 19. It consists primarily of small and medium raindrops.

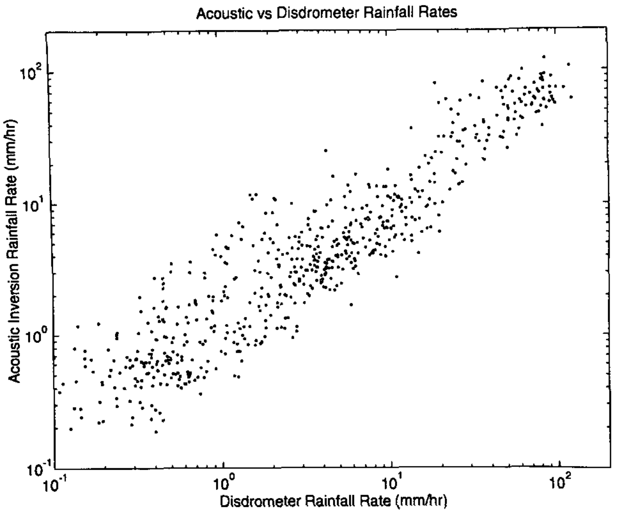


FIG. 12. One-minute rainfall rate estimates using the disdrometer and the acoustical inversion technique. These data are from six subtropical convective systems recorded at Miami. The correlation coefficient is $r = 0.90$.

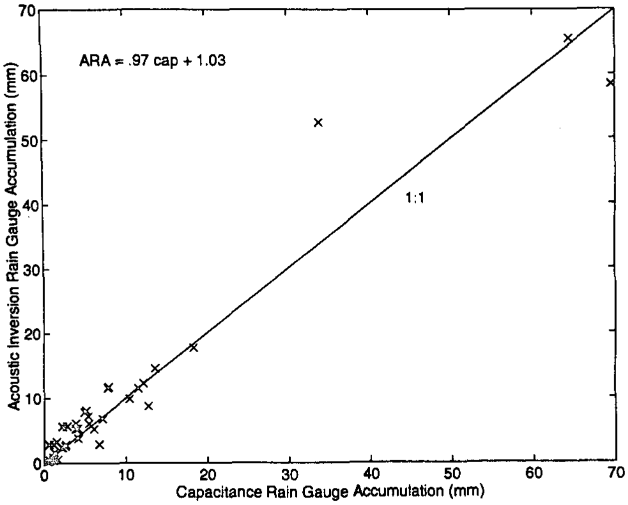


FIG. 13. Rainfall accumulation totals for 40 individual rain events using the capacitance rain gauge and the acoustic inversion (ARA) technique. The correlation coefficient is $r = 0.97$. The first-order linear regression equation is shown in the figure.

The acoustic inversion for another long and heavy rainfall event (over 60-mm accumulation) is shown in Fig. 11. Between minutes 50 and 100, several bands of extremely heavy rainfall containing very large raindrops occurred and were acoustically detected. Equally well detected were the surges and lulls in rainfall before minute 50 and after minute 100. Note, in particular, the lulls at minutes 115, 170, and 225. The ability of ARA inversion to accurately estimate the small drop populations is unexpected as the physical mechanism for small drop sound production is known to be sensitive to wind (Nystuen 1993; Medwin et al. 1989). At the AOML site, wind speeds were generally below 5 m $s^{-1}$. At a more exposed site, estimates of small drop populations may be poorer.

### b. *Rainfall products using acoustical rainfall analysis (ARA)*

Once the drop size distribution has been estimated, any of several rainfall products can be estimated. These include rainfall rate, liquid water content, optical extinction, reflectivity, etc. These quantities are all moments of the drop size distribution; that is,

$$M_x = \int D^x \mathrm{dsd}(D) dD.$$

For example, $M_0$ is the number of drops per unit volume, $M_2$ is proportional to the cross-sectional area of the rain ($mm^2$ $m^{-3}$), $M_3$ is proportional to the liquid water volume, $M_{3.6}$ is proportional to $R$, the rainfall rate, and $M_6$ is reflectivity (the quantity measured by weather radar).

Figure 12 shows a comparison of 1-min rainfall rates during six subtropical convective systems estimation using both the acoustic method and the disdrometer. Overall, the agreement is good (correlation coefficient, $r = 0.90$), although some scatter is observed. This is likely due to timing errors (50-m separation between the hydrophone and the disdrometer) or to the relative insensitivity of the ARA technique to the medium raindrops. These raindrops are relatively quiet and yet occasionally comprise a significant portion of the drop size distribution. Further algorithm development may

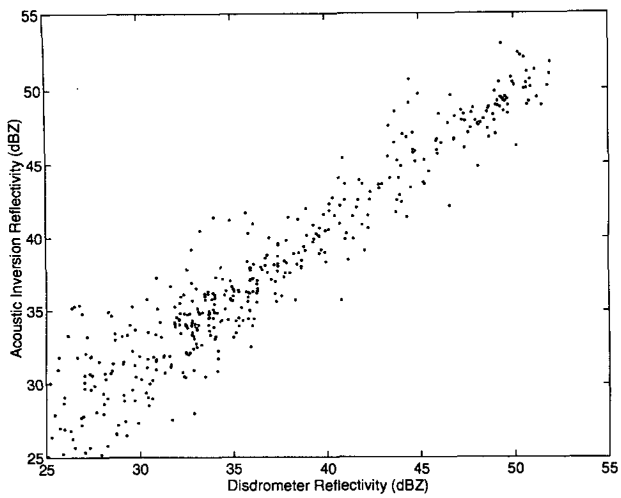


FIG. 14. One-minute rainfall reflectivity estimates using the disdrometer and the acoustical inversion technique. Rainfall reflectivity is the rainfall parameter measured by weather radars. These data are from six subtropical convective systems recorded at Miami. The correlation coefficient is $r = 0.93$.

improve the rainfall rate estimation. Accumulation totals (Fig. 13) are in better agreement with the other rain gauges. For these 40 events, the correlation coefficient is $r = 0.97$.

One significant advantage of the acoustical rainfall measurement is the ability to estimate rainfall parameters other than rainfall rate. Figure 14 shows that the acoustical estimates of electromagnetic reflectivity are in excellent agreement with the disdrometer reflectivity estimates. This is, in part, because the reflectivity is also relatively insensitive to medium and small raindrops (reflectivity is the sixth moment of the drop size distribution). Because weather radars measure reflectivity, radar estimates of rainfall rate (a different moment of the drop size distribution than reflectivity) are often erroneous (Atlas and Ulbrich 1977). The acoustical rainfall measurement provides a link between reflectivity and rainfall rate that should be of use in studies of radar rainfall estimates.

## 5. Conclusions

Acoustical rainfall analysis (ARA) uses the naturally occurring underwater sound generated by precipitation to detect, classify, and quantify rainfall. It is a rainfall measurement technique that is inherently applicable to oceanic regions and can be adapted for use in lakes and reservoirs. Currently, few in situ measurement techniques are available for oceanic regions, and yet rainfall estimates are needed for a variety of oceanographic, meteorological, and climatic studies.

Using laboratory studies of the sound produced by single drops for guidance, a transformation matrix for the formal inversion of the underwater sound field is presented. Both the forward problem, predicting the sound field from the drop size distribution, and the inverse problem, predicting the drop size distribution from the sound field, are demonstrated. Although only four drop size populations are estimated using the ARA inversion, most of the features of the drop size distribution are accurately obtained. From the drop size distribution estimates, a variety of rainfall parameters—for example, rainfall rate, reflectivity, liquid water content, etc.—can be calculated. Furthermore, the acoustic signal can be used to identify the convective and stratiform portions of the rainfall.

The ARA rainfall measurement technique was applied to 40 rain events recorded in Miami, Florida. Accumulation totals are in excellent agreement with other automatic rain gauges (correlation coefficient $r = 0.97$; linear regression slope = 0.97). One-minute rainfall rate estimates, from six subtropical convective events, are also highly correlated to disdrometer rainfall rate estimates ($r = 0.90$).

The acoustic rainfall measurement is relatively insensitive to medium-size raindrops (1.2–2.4-mm diameter) as these raindrops do not produce significant sound underwater. On the other hand, the sound field is dominated by the very large raindrops (>4.1 mm diameter) when they are present. The sensitivity of the sound field to very large raindrops results in a very good estimate of reflectivity (the rainfall parameter measured by radars). ARA reflectivity values are within 3 dB of the disdrometer estimates over a wide range of reflectivity (correlation coefficient $r = 0.93$). The ability to acoustically estimate both rainfall rate and reflectivity should be of use in radar rainfall studies.

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